

OVERVIEW OF THE SUPPLEMENTAL MATERIAL

Our supplemental material consists of four main sections:

- **Detailed Proofs:** proofs of Theorem 1, Theorem 2, Theorem 3, Theorem 4, Section III-E1, Theorem 5, and Theorem 6.
- **Complexity Analysis:** space complexity analysis of OLEC-SB.
- **Additional Experiments:** supplementary experimental results, including dataset details, compared methods, clustering performance (ARI/F-score), and statistical analysis.
- **Notation Summary:** main notations used in OLEC-SB and the theoretical analysis.

SUPPLEMENTAL MATERIAL A DETAILED PROOF

In this section, we provide detailed proofs for the theoretical results presented in the main paper, including Theorem 1, Theorem 2, Theorem 3, Theorem 4, Section III-E1, Theorem 5, and Theorem 6.

A. Proof of Theorem 1

Proof. By the Cauchy–Schwarz inequality,

$$\left(\sum_{l=1}^K |\mathcal{C}_l|^2 \right) \left(\sum_{l=1}^K 1^2 \right) \geq \left(\sum_{l=1}^K |\mathcal{C}_l| \cdot 1 \right)^2. \quad (\text{S1})$$

Since $\sum_{l=1}^K 1^2 = K$ and $\sum_{l=1}^K |\mathcal{C}_l| = n$, we obtain

$$\sum_{l=1}^K |\mathcal{C}_l|^2 \geq \frac{n^2}{K}, \quad (\text{S2})$$

and the equality holds if and only if $|\mathcal{C}_1| = \dots = |\mathcal{C}_K| = n/K$. This completes the proof of Theorem 1. $\square$

B. Proof of Theorem 2

Proof. By Eq. (1), since $\bar{\mathcal{C}}_l = \bigcup_{h \neq l} \mathcal{C}_h$ and the clusters are pairwise disjoint, the similarity between $\mathcal{C}_l$ and its complement $\bar{\mathcal{C}}_l$ can be decomposed as the sum of the similarities between $\mathcal{C}_l$ and each of the remaining clusters. Thus, we have

$$\text{sim}_k(\mathcal{C}_l, \bar{\mathcal{C}}_l) = \sum_{h \neq l} \text{sim}_k(\mathcal{C}_l, \mathcal{C}_h) = \sum_{i \in \mathcal{C}_l} \sum_{j \notin \mathcal{C}_l} w_{ij}^k. \quad (\text{S3})$$

On the other hand, let $\mathbf{y}_l$ denote the indicator vector of cluster $\mathcal{C}_l$, i.e., $(\mathbf{y}_l)_i = 1$ if $x_i \in \mathcal{C}_l$, and $(\mathbf{y}_l)_i = 0$ otherwise. Since $\mathbf{D}_k$ is the diagonal degree matrix associated with $\mathbf{W}_k$, the quadratic form $\mathbf{y}_l^\top \mathbf{D}_k \mathbf{y}_l$ selects precisely the degree terms corresponding to the samples in $\mathcal{C}_l$. Thus,

$$\mathbf{y}_l^\top \mathbf{D}_k \mathbf{y}_l = \sum_{i=1}^n (\mathbf{y}_l)_i^2 d_i^k = \sum_{i \in \mathcal{C}_l} d_i^k = \sum_{i \in \mathcal{C}_l} \sum_{j=1}^n w_{ij}^k. \quad (\text{S4})$$

Similarly,

$$\mathbf{y}_l^\top \mathbf{W}_k \mathbf{y}_l = \sum_{i=1}^n \sum_{j=1}^n (\mathbf{y}_l)_i w_{ij}^k (\mathbf{y}_l)_j = \sum_{i \in \mathcal{C}_l} \sum_{j \in \mathcal{C}_l} w_{ij}^k. \quad (\text{S5})$$

Therefore,

$$\mathbf{y}_l^\top (\mathbf{D}_k - \mathbf{W}_k) \mathbf{y}_l = \sum_{i \in \mathcal{C}_l} \sum_{j=1}^n w_{ij}^k - \sum_{i \in \mathcal{C}_l} \sum_{j \in \mathcal{C}_l} w_{ij}^k = \sum_{i \in \mathcal{C}_l} \sum_{j \notin \mathcal{C}_l} w_{ij}^k. \quad (\text{S6})$$

Combining the above equality with the definition of inter-cluster similarity yields

$$\sum_{h \neq l} \text{sim}_k(\mathcal{C}_l, \mathcal{C}_h) = \mathbf{y}_l^\top (\mathbf{D}_k - \mathbf{W}_k) \mathbf{y}_l. \quad (\text{S7})$$

This completes the proof of Theorem 2. $\square$

C. Proof of Theorem 3

Proof. Since $\Delta_k = \mathbf{A}_k^\top \mathbf{A}_k$ is diagonal with strictly positive diagonal entries, Δ_k^{-1} exists. For any sample $\mathbf{x}_i$, denote by $r(i)$ the cluster index such that $(\mathbf{A}_k)_{i r(i)} = 1$. From Eq. (5), the (i, j) -th entry of $\mathbf{W}_k$ can be written as

$$\begin{aligned} (\mathbf{W}_k)_{ij} &= \sum_{r=1}^{l_k} (\mathbf{A}_k)_{ir} (\Delta_k^{-1})_{rr} (\mathbf{A}_k)_{jr} \\ &= \begin{cases} 1/|\mathcal{C}_{r(i)}^k|, & \text{if } \mathbf{x}_j \in \mathcal{C}_{r(i)}^k, \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (\text{S8})$$

where $|\mathcal{C}_{r(i)}^k|$ denotes the size of the cluster containing $\mathbf{x}_i$ under π_k . Therefore, the i -th row sum is

$$\sum_{j=1}^n (\mathbf{W}_k)_{ij} = \sum_{j \in \mathcal{C}_{r(i)}^k} \frac{1}{|\mathcal{C}_{r(i)}^k|} = 1, \quad (\text{S9})$$

which implies $\mathbf{W}_k \mathbf{1}_n = \mathbf{1}_n$. Moreover, $\mathbf{W}_k$ is symmetric because Δ_k^{-1} is diagonal and $\mathbf{W}_k = \mathbf{A}_k \Delta_k^{-1} \mathbf{A}_k^\top$, hence $\mathbf{1}_n^\top \mathbf{W}_k = (\mathbf{W}_k \mathbf{1}_n)^\top = \mathbf{1}_n^\top$. Thus, $\mathbf{W}_k$ is doubly stochastic. Finally, by definition $(\mathbf{D}_k)_{ii} = \sum_{j=1}^n w_{ij}^k$, we obtain $(\mathbf{D}_k)_{ii} = 1$ for all i , i.e., $\mathbf{D}_k = \mathbf{I}$. This completes the proof of Theorem 3. $\square$

D. Proof of Theorem 4

Proof. Since $\mathbf{W}_k = \mathbf{B}_k \mathbf{B}_k^\top$ is doubly stochastic (by Theorem 3), it holds that $\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top \succeq \mathbf{0}$, and thus $\text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top) \mathbf{Y}) \geq 0$ for all k . For $\gamma > 1$, problem (21) is convex in α over the probability simplex, so the KKT conditions are sufficient for global optimality.

By introducing the Lagrange multipliers β for $\mathbf{1}_m^\top \alpha = 1$ and $\mu_k \geq 0$ for $\alpha_k \geq 0$, the Lagrangian function is given by

$$\begin{aligned} \mathcal{L}(\alpha, \beta, \mu) &= \sum_{k=1}^m \alpha_k^\gamma \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top) \mathbf{Y}) - \beta (\mathbf{1}_m^\top \alpha - 1) - \mu^\top \alpha. \end{aligned} \quad (\text{S10})$$

Taking the derivative of $\mathcal{L}$ w.r.t. α_k and setting it to zero yields

$$\frac{\partial \mathcal{L}}{\partial \alpha_k} = \gamma \alpha_k^{\gamma-1} \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top) \mathbf{Y}) - \beta - \mu_k = 0. \quad (\text{S11})$$

Because $\sum_{k=1}^m \alpha_k = 1$, there exists j such that $\alpha_j > 0$, which implies $\mu_j = 0$ and thus

$$\beta = \gamma \alpha_j^{\gamma-1} \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_j \mathbf{B}_j^\top) \mathbf{Y}) > 0. \quad (\text{S12})$$

This further indicates that $\alpha_k = 0$ cannot be optimal: if $\alpha_k = 0$ for some k , then Eq. (S11) reduces to $-\beta - \mu_k = 0$, which yields $\mu_k = -\beta < 0$ and violates $\mu_k \geq 0$. Therefore, the optimum satisfies $\alpha_k > 0$ for all k , and hence $\mu_k = 0$ for all k . Substituting $\mu_k = 0$ into Eq. (S11) yields

$$\gamma \alpha_k^{\gamma-1} \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top) \mathbf{Y}) = \beta, \quad (\text{S13})$$

which leads to

$$\alpha_k = \left(\frac{\beta}{\gamma \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_k \mathbf{B}_k^\top) \mathbf{Y})} \right)^{\frac{1}{\gamma-1}}. \quad (\text{S14})$$

Finally, enforcing $\sum_{k=1}^m \alpha_k = 1$ eliminates β and yields the closed-form update in Eq. (22). This completes the proof of Theorem 4. $\square$

E. Proof of Theorem III-E1

Proof. Let $\mathcal{L}(\mathbf{Y}, \boldsymbol{\alpha})$ denote the objective in problem (6). In t -th iteration, given $(\mathbf{Y}^{(t)}, \boldsymbol{\alpha}^{(t)})$, Algorithm 1 updates $\mathbf{Y}$ and $\boldsymbol{\alpha}$ by alternately solving the two corresponding subproblems.

Update $\mathbf{Y}$. With $\boldsymbol{\alpha}^{(t)}$ fixed, the $\mathbf{Y}$ -subproblem is solved to obtain $\mathbf{Y}^{(t+1)}$. By optimality,

$$\mathcal{L}(\mathbf{Y}^{(t+1)}, \boldsymbol{\alpha}^{(t)}) \leq \mathcal{L}(\mathbf{Y}^{(t)}, \boldsymbol{\alpha}^{(t)}). \quad (\text{S15})$$

Update $\boldsymbol{\alpha}$. With $\mathbf{Y}^{(t+1)}$ fixed, the $\boldsymbol{\alpha}$ -subproblem is solved to obtain $\boldsymbol{\alpha}^{(t+1)}$. Similarly,

$$\mathcal{L}(\mathbf{Y}^{(t+1)}, \boldsymbol{\alpha}^{(t+1)}) \leq \mathcal{L}(\mathbf{Y}^{(t+1)}, \boldsymbol{\alpha}^{(t)}). \quad (\text{S16})$$

Combining (S15) and (S16) yields the monotonic descent property

$$\mathcal{L}(\mathbf{Y}^{(t+1)}, \boldsymbol{\alpha}^{(t+1)}) \leq \mathcal{L}(\mathbf{Y}^{(t)}, \boldsymbol{\alpha}^{(t)}). \quad (\text{S17})$$

Moreover, $\boldsymbol{\alpha}$ is restricted to the probability simplex $\{\boldsymbol{\alpha} \succeq \mathbf{0} \mid \mathbf{1}_m^\top \boldsymbol{\alpha} = 1\}$ and $\mathbf{Y} \in \text{Ind}$ by construction; hence the sequence $\{\mathcal{L}(\mathbf{Y}^{(t)}, \boldsymbol{\alpha}^{(t)})\}$ is bounded below. Thus, Algorithm 1 converges. This completes the proof of Theorem III-E1. $\square$

F. Proof of Theorem 5

Proof. By Eq. (12), the row-wise update of $\mathbf{x}_i$ is determined by minimizing $\varphi(s)$ over $s \in \{1, \dots, K\}$, where $\varphi(s)$ denotes the objective difference between assigning $\mathbf{x}_i$ to the s -th cluster and the auxiliary zero-row state.

Since $\mathbf{Y}^{(s)}$ and $\mathbf{Y}^{(0)}$ differ only in the assignment of $\mathbf{x}_i$, only the s -th column changes. Therefore, the balancing term satisfies

$$\begin{aligned} (\mathbf{y}_s^{(s)})^\top \mathbf{1}_n \mathbf{1}_n^\top \mathbf{y}_s^{(s)} - (\mathbf{y}_s^{(0)})^\top \mathbf{1}_n \mathbf{1}_n^\top \mathbf{y}_s^{(0)} \\ = 2(\mathbf{y}_s^{(0)})^\top \mathbf{1}_n + 1 = 2|\mathcal{C}_s^{-i}| + 1. \end{aligned} \quad (\text{S18})$$

Similarly, for each k , expanding the quadratic form gives

$$\begin{aligned} (\mathbf{y}_s^{(s)})^\top \mathbf{B}_k \mathbf{B}_k^\top \mathbf{y}_s^{(s)} - (\mathbf{y}_s^{(0)})^\top \mathbf{B}_k \mathbf{B}_k^\top \mathbf{y}_s^{(0)} \\ = 2(\mathbf{y}_s^{(0)})^\top \mathbf{B}_k (\mathbf{b}_k)_i^\top + (\mathbf{b}_k)_i (\mathbf{b}_k)_i^\top. \end{aligned} \quad (\text{S19})$$

Substituting Eqs (S18)-(S19) into $\varphi(s)$, we obtain

$$\begin{aligned} \varphi(s) = -2 \left(\sum_{k=1}^m \alpha_k^\gamma (\mathbf{y}_s^{(0)})^\top \mathbf{B}_k (\mathbf{b}_k)_i^\top - \lambda |\mathcal{C}_s^{-i}| \right) \\ + \lambda - \sum_{k=1}^m \alpha_k^\gamma (\mathbf{b}_k)_i (\mathbf{b}_k)_i^\top. \end{aligned} \quad (\text{S20})$$

The last two terms are independent of s . Hence, minimizing $\varphi(s)$ is equivalent to maximizing

$$\sum_{k=1}^m \alpha_k^\gamma (\mathbf{y}_s^{(0)})^\top \mathbf{B}_k (\mathbf{b}_k)_i^\top - \lambda |\mathcal{C}_s^{-i}|. \quad (\text{S21})$$

Next, using $\mathbf{W}_k = \mathbf{B}_k \mathbf{B}_k^\top$, we have

$$\sum_{k=1}^m \alpha_k^\gamma (\mathbf{y}_s^{(0)})^\top \mathbf{B}_k (\mathbf{b}_k)_i^\top = \sum_{k=1}^m \alpha_k^\gamma \sum_{j \in \mathcal{C}_s^{-i}} (\mathbf{W}_k)_{ji} = |\mathcal{C}_s^{-i}| \bar{a}_{is}, \quad (\text{S22})$$

where

$$\bar{a}_{is} = \frac{1}{|\mathcal{C}_s^{-i}|} \sum_{k=1}^m \alpha_k^\gamma \sum_{j \in \mathcal{C}_s^{-i}} (\mathbf{W}_k)_{ji}. \quad (\text{S23})$$

Hence, the row-wise update of $\mathbf{x}_i$ is

$$q = \arg \max_{s \in \{1, \dots, K\}} (|\mathcal{C}_s^{-i}| \bar{a}_{is} - \lambda |\mathcal{C}_s^{-i}|). \quad (\text{S24})$$

This completes the proof of Theorem 5. $\square$

G. Proof of Theorem 6

Proof. To prove Theorem 6, we need the following lemmas 1-3.

Lemma 1. *The perturbation of the weighted normalized co-association term satisfies*

$$\begin{aligned} \sup_{\mathbf{x}_i, \mathbf{x}_j} \left| \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right| \\ \leq c_B \frac{\sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}}{n} \bar{\Delta}_B, \end{aligned} \quad (\text{S25})$$

where $c_B \geq \sqrt{K}$ is the base-clustering perturbation constant.

Proof. Let $\mathbf{P}_{k,\mathcal{X}} = \mathbf{B}_{k,\mathcal{X}} \mathbf{B}_{k,\mathcal{X}}^\top$ and $\mathbf{P}_{k,\mathcal{X}^{(r)}} = \mathbf{B}_{k,\mathcal{X}^{(r)}} \mathbf{B}_{k,\mathcal{X}^{(r)}}^\top$. Since $\mathbf{B}_{k,\mathcal{X}^{(r)}} = \mathbf{B}_{k,\mathcal{X}} + \Delta \mathbf{B}_k$, we have

$$\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}} = \mathbf{B}_{k,\mathcal{X}} \Delta \mathbf{B}_k^\top + \Delta \mathbf{B}_k \mathbf{B}_{k,\mathcal{X}}^\top + \Delta \mathbf{B}_k \Delta \mathbf{B}_k^\top. \quad (\text{S26})$$

Because $\mathbf{B}_{k,\mathcal{X}}$ is a normalized indicator matrix, $\mathbf{B}_{k,\mathcal{X}}^\top \mathbf{B}_{k,\mathcal{X}} = \mathbf{I}$, and thus $\|\mathbf{B}_{k,\mathcal{X}}\|_2 = 1$. Therefore,

$$\|\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}}\|_F \leq 2\|\Delta \mathbf{B}_k\|_F + \|\Delta \mathbf{B}_k\|_F^2 \leq \bar{\Delta}_B. \quad (\text{S27})$$

For any pair $(\mathbf{x}_i, \mathbf{x}_j)$, $p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j)$ is an entry of $\mathbf{P}_{k,\mathcal{X}}$. Hence,

$$|p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j)| \leq \|\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}}\|_F \leq \bar{\Delta}_B. \quad (\text{S28})$$

Since $\gamma > 1$, we have $\sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma \leq \sum_{k=1}^m \alpha_{k,\mathcal{X}} = 1$. Therefore,

$$\begin{aligned} & \sup_{\mathbf{x}_i, \mathbf{x}_j} \left| \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - \sum_{k=1}^m \alpha_{k,\mathcal{X}^{(r)}}^\gamma p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right| \\ & \leq \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma \sup_{\mathbf{x}_i, \mathbf{x}_j} |p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j)| \leq \bar{\Delta}_B. \end{aligned} \quad (\text{S29})$$

Moreover, since $\sum_{l=1}^K |\mathcal{C}_l| = n$, the Cauchy-Schwarz inequality gives $\sum_{l=1}^K |\mathcal{C}_l|^2 \geq n^2/K$. Thus, $1 \leq \sqrt{K} \sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2/n}$. Taking $c_B \geq \sqrt{K}$, we obtain the desired bound. This completes the proof of Lemma 1. $\square$

Lemma 2. For fixed $\mathbf{Y}$, let $d_{k,\mathcal{X}} = \text{tr}(\mathbf{Y}^\top (\mathbf{I} - \mathbf{B}_{k,\mathcal{X}} \mathbf{B}_{k,\mathcal{X}}^\top) \mathbf{Y})$ be the base-clustering inconsistency in the k -th base clustering, and define $d_{k,\mathcal{X}^{(r)}}$ analogously. Suppose that $d_{k,\mathcal{X}}$ and $d_{k,\mathcal{X}^{(r)}}$ satisfy $nd_{\min} \leq d_{k,\mathcal{X}}, d_{k,\mathcal{X}^{(r)}} \leq nd_{\max}$, where $0 < d_{\min} \leq d_{\max} < \infty$. Then the adaptive weighting strategy satisfies

$$\|\alpha_{\mathcal{X}} - \alpha_{\mathcal{X}^{(r)}}\|_\infty \leq c_\alpha \frac{\sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}}{n} \bar{\Delta}_B, \quad (\text{S30})$$

where $c_\alpha > 0$ depends on γ , m , $d_{\min}$, and $d_{\max}$.

Proof. Let $\mathbf{d}_{\mathcal{X}} = (d_{1,\mathcal{X}}, \dots, d_{m,\mathcal{X}})$ and $\mathbf{d}_{\mathcal{X}^{(r)}} = (d_{1,\mathcal{X}^{(r)}}, \dots, d_{m,\mathcal{X}^{(r)}})$. For $a = 1/(\gamma - 1) > 0$, define $\Phi_k(\mathbf{d}) = d_k^{-a} / \sum_{r=1}^m d_r^{-a}$, such that $\alpha_{k,\mathcal{X}} = \Phi_k(\mathbf{d}_{\mathcal{X}})$ and $\alpha_{k,\mathcal{X}^{(r)}} = \Phi_k(\mathbf{d}_{\mathcal{X}^{(r)}})$.

The partial derivative of $\Phi_k(\mathbf{d})$ with respect to d_j is

$$\frac{\partial \Phi_k}{\partial d_j} = -a \mathbb{I}[j = k] \frac{d_k^{-a-1}}{\sum_{r=1}^m d_r^{-a}} + a \frac{d_k^{-a} d_j^{-a-1}}{(\sum_{r=1}^m d_r^{-a})^2}. \quad (\text{S31})$$

Since $nd_{\min} \leq d_j \leq nd_{\max}$, all partial derivatives are bounded by a constant of order $1/n$. Hence, there exists $c_\alpha > 0$, independent of n , such that $|\partial \Phi_k / \partial d_j| \leq c_\alpha/n$ for all k, j . By the mean value theorem, with the factor depending on m absorbed into c_α , we have

$$\|\alpha_{\mathcal{X}} - \alpha_{\mathcal{X}^{(r)}}\|_\infty \leq \frac{c_\alpha}{n} \max_k |d_{k,\mathcal{X}} - d_{k,\mathcal{X}^{(r)}}|. \quad (\text{S32})$$

To control $|d_{k,\mathcal{X}} - d_{k,\mathcal{X}^{(r)}}|$, let $\mathbf{P}_{k,\mathcal{X}} = \mathbf{B}_{k,\mathcal{X}} \mathbf{B}_{k,\mathcal{X}}^\top$ and $\mathbf{P}_{k,\mathcal{X}^{(r)}} = \mathbf{B}_{k,\mathcal{X}^{(r)}} \mathbf{B}_{k,\mathcal{X}^{(r)}}^\top$. Then

$$\begin{aligned} |d_{k,\mathcal{X}} - d_{k,\mathcal{X}^{(r)}}| &= |\text{tr}(\mathbf{Y} \mathbf{Y}^\top (\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}}))| \\ &\leq \|\mathbf{Y} \mathbf{Y}^\top\|_F \|\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}}\|_F. \end{aligned} \quad (\text{S33})$$

Since $\mathbf{Y} \in \text{Ind}$, we have $\|\mathbf{Y} \mathbf{Y}^\top\|_F = \sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}$. From Lemma 1, $\|\mathbf{P}_{k,\mathcal{X}^{(r)}} - \mathbf{P}_{k,\mathcal{X}}\|_F \leq \bar{\Delta}_B$, which implies

$$\max_k |d_{k,\mathcal{X}} - d_{k,\mathcal{X}^{(r)}}| \leq \sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2} \bar{\Delta}_B. \quad (\text{S34})$$

Substituting Eq. (S34) into Eq. (S32) yields the desired inequality. This completes the proof of Lemma 2. $\square$

Lemma 3. Let q be the cluster index assigned to $\mathbf{x}_i$ before perturbation. For any candidate cluster index $s \neq q$, define

$$\mathcal{M}_{i,s}^{(\lambda)} = \frac{|\mathcal{C}_q^{-i}| \bar{a}_{iq} - |\mathcal{C}_s^{-i}| \bar{a}_{is} + \lambda (|\mathcal{C}_s^{-i}| - |\mathcal{C}_q^{-i}|)}{\sqrt{|\mathcal{C}_q^{-i}|} + \sqrt{|\mathcal{C}_s^{-i}|} + 2m\gamma c_\alpha \frac{\sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}}{n}}. \quad (\text{S35})$$

Let $\mathcal{M}_\lambda = \min_{1 \leq i \leq n} \min_{1 \leq s \leq K, s \neq q} \mathcal{M}_{i,s}^{(\lambda)}$. If $\bar{\Delta}_B < \mathcal{M}_\lambda$, then all cluster assignments remain unchanged under the base-clustering perturbation from $\mathcal{X}$ to $\mathcal{X}^{(r)}$. Equivalently,

$$\|\mathbb{I}[g_{\mathcal{X}}(\mathbf{x}_i) = g_{\mathcal{X}}(\mathbf{x}_j)] - \mathbb{I}[g_{\mathcal{X}^{(r)}}(\mathbf{x}_i) = g_{\mathcal{X}^{(r)}}(\mathbf{x}_j)]\| = 0. \quad (\text{S36})$$

Proof. By Theorem 5, the row-wise update of $\mathbf{x}_i$ is determined by $q = \arg \max_{s \in \{1, \dots, K\}} (|\mathcal{C}_s^{-i}| \bar{a}_{is} - \lambda |\mathcal{C}_s^{-i}|)$. Thus, for any $s \neq q$, the difference between the cluster assignment criterion for index q and that for index s before perturbation is

$$|\mathcal{C}_q^{-i}| \bar{a}_{iq} - |\mathcal{C}_s^{-i}| \bar{a}_{is} + \lambda (|\mathcal{C}_s^{-i}| - |\mathcal{C}_q^{-i}|). \quad (\text{S37})$$

After perturbing, denote the i -th row of $\Delta \mathbf{B}_k$ by $(\Delta \mathbf{b}_k)_i$, then $(\mathbf{b}_{k,\mathcal{X}^{(r)}})_i = (\mathbf{b}_{k,\mathcal{X}})_i + (\Delta \mathbf{b}_k)_i$. Under the perturbed dataset $\mathcal{X}^{(r)}$, define $\bar{a}_{is}^{(r)} = 1/|\mathcal{C}_s^{-i}| \sum_{k=1}^m \alpha_{k,\mathcal{X}^{(r)}}^\gamma (\mathbf{y}_s^{(0)})^\top \mathbf{B}_{k,\mathcal{X}^{(r)}} (\mathbf{b}_{k,\mathcal{X}^{(r)}})_i^\top$. Then,

$$\begin{aligned} & |\mathcal{C}_s^{-i}| \bar{a}_{is}^{(r)} - |\mathcal{C}_s^{-i}| \bar{a}_{is} \\ &= \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma (\mathbf{y}_s^{(0)})^\top (\mathbf{B}_{k,\mathcal{X}} (\Delta \mathbf{b}_k)_i^\top + \Delta \mathbf{B}_k (\mathbf{b}_{k,\mathcal{X}})_i^\top + \Delta \mathbf{B}_k (\Delta \mathbf{b}_k)_i^\top) \\ &+ \sum_{k=1}^m (\alpha_{k,\mathcal{X}^{(r)}}^\gamma - \alpha_{k,\mathcal{X}}^\gamma) (\mathbf{y}_s^{(0)})^\top \mathbf{B}_{k,\mathcal{X}^{(r)}} (\mathbf{b}_{k,\mathcal{X}^{(r)}})_i^\top. \end{aligned} \quad (\text{S38})$$

Since $\|\mathbf{y}_s^{(0)}\|_2 = \sqrt{|\mathcal{C}_s^{-i}|}$, $\|(\mathbf{b}_{k,\mathcal{X}})_i\|_2 \leq 1$, $\|(\Delta \mathbf{b}_k)_i\|_2 \leq \|\Delta \mathbf{B}_k\|_F$, and $\sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma \leq 1$, the first term is bounded by $\sqrt{|\mathcal{C}_s^{-i}|} \bar{\Delta}_B$. For the second term, since $0 \leq \alpha_{k,\mathcal{X}}, \alpha_{k,\mathcal{X}^{(r)}} \leq 1$ and $\gamma > 1$, we have $|\alpha_{k,\mathcal{X}^{(r)}}^\gamma - \alpha_{k,\mathcal{X}}^\gamma| \leq \gamma |\alpha_{k,\mathcal{X}^{(r)}} - \alpha_{k,\mathcal{X}}|$. By Lemma 2, $\|\alpha_{\mathcal{X}^{(r)}} - \alpha_{\mathcal{X}}\|_\infty \leq c_\alpha \sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2} \bar{\Delta}_B/n$. Moreover, since $\mathbf{B}_{k,\mathcal{X}^{(r)}} \mathbf{B}_{k,\mathcal{X}^{(r)}}^\top$ is nonnegative and doubly stochastic, $(\mathbf{y}_s^{(0)})^\top \mathbf{B}_{k,\mathcal{X}^{(r)}} (\mathbf{b}_{k,\mathcal{X}^{(r)}})_i^\top \leq 1$. Therefore, the second term is bounded by $m\gamma c_\alpha \sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2} \bar{\Delta}_B/n$.

Combining the above bounds, we obtain

$$|\mathcal{C}_s^{-i}| \bar{a}_{is}^{(r)} - |\mathcal{C}_s^{-i}| \bar{a}_{is} \leq \left(\sqrt{|\mathcal{C}_s^{-i}|} + m\gamma c_\alpha \frac{\sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}}{n} \right) \bar{\Delta}_B. \quad (\text{S39})$$

Similarly,

$$|\mathcal{C}_q^{-i}| \bar{a}_{iq}^{(r)} - |\mathcal{C}_q^{-i}| \bar{a}_{iq} \leq \left(\sqrt{|\mathcal{C}_q^{-i}|} + m\gamma c_\alpha \frac{\sqrt{\sum_{l=1}^K |\mathcal{C}_l|^2}}{n} \right) \bar{\Delta}_B. \quad (\text{S40})$$

Therefore, the perturbed difference between the cluster assignment criteria of indices q and s satisfies

$$\begin{aligned} & \left(|C_q^{-i}| \bar{a}_{iq}^{(r)} - \lambda |C_q^{-i}| \right) - \left(|C_s^{-i}| \bar{a}_{is}^{(r)} - \lambda |C_s^{-i}| \right) \\ & \geq |C_q^{-i}| \bar{a}_{iq} - |C_s^{-i}| \bar{a}_{is} + \lambda (|C_s^{-i}| - |C_q^{-i}|) \\ & \quad - \left(\sqrt{|C_q^{-i}|} + \sqrt{|C_s^{-i}|} + 2m\gamma c_\alpha \frac{\sqrt{\sum_{l=1}^K |C_l|^2}}{n} \right) \bar{\Delta}_B. \end{aligned} \quad (\text{S41})$$

Thus, if $\bar{\Delta}_B < \mathcal{M}_{i,s}^{(\lambda)}$, the above lower bound remains positive. Taking the minimum over all i and $s \neq q$, the condition $\bar{\Delta}_B < \mathcal{M}_\lambda$ ensures that all cluster assignments remain unchanged. This completes the proof of Lemma 3. $\square$

Now, we complete the proof of Theorem 6

We consider two cases. First, suppose $\bar{\Delta}_B < \mathcal{M}_\lambda$. By Lemma 3, all cluster assignments remain unchanged. Thus, for any pair $(\mathbf{x}_i, \mathbf{x}_j)$, the indicator $\mathbb{I}[g_{\mathcal{X}}(\mathbf{x}_i) = g_{\mathcal{X}}(\mathbf{x}_j)]$ remains unchanged after replacing $\mathcal{X}$ by $\mathcal{X}^{(r)}$. Hence, the difference between the two pairwise losses is only caused by the perturbation of the weighted normalized co-association term and the adaptive weights:

$$\begin{aligned} & \sup_{\mathbf{x}_i, \mathbf{x}_j} \left| \ell_\lambda(\mathbf{x}_i, \mathbf{x}_j; A_\lambda(\mathcal{X})) - \ell_\lambda(\mathbf{x}_i, \mathbf{x}_j; A_\lambda(\mathcal{X}^{(r)})) \right| \\ & \leq \left| \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - \sum_{k=1}^m \alpha_{k,\mathcal{X}^{(r)}}^\gamma p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right|. \end{aligned} \quad (\text{S42})$$

The right-hand side can be decomposed as

$$\begin{aligned} & \left| \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - \sum_{k=1}^m \alpha_{k,\mathcal{X}^{(r)}}^\gamma p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right| \\ & \leq \left| \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma (p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) - p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j)) \right| \\ & \quad + \left| \sum_{k=1}^m (\alpha_{k,\mathcal{X}}^\gamma - \alpha_{k,\mathcal{X}^{(r)}}^\gamma) p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right|. \end{aligned} \quad (\text{S43})$$

By Lemma 1, the first term is bounded by $c_B \sqrt{\sum_{l=1}^K |C_l|^2 \bar{\Delta}_B / n}$. For the second term, since $\alpha_{k,\mathcal{X}}, \alpha_{k,\mathcal{X}^{(r)}} \in [0, 1]$ and $\gamma > 1$, the mean value theorem gives $|\alpha_{k,\mathcal{X}}^\gamma - \alpha_{k,\mathcal{X}^{(r)}}^\gamma| \leq \gamma |\alpha_{k,\mathcal{X}} - \alpha_{k,\mathcal{X}^{(r)}}|$. Together with $0 \leq p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \leq 1$, we have

$$\left| \sum_{k=1}^m (\alpha_{k,\mathcal{X}}^\gamma - \alpha_{k,\mathcal{X}^{(r)}}^\gamma) p_{k,\mathcal{X}^{(r)}}(\mathbf{x}_i, \mathbf{x}_j) \right| \leq m\gamma \|\alpha_{\mathcal{X}} - \alpha_{\mathcal{X}^{(r)}}\|_\infty. \quad (\text{S44})$$

By Lemma 2, the second term is bounded by $m\gamma c_\alpha \sqrt{\sum_{l=1}^K |C_l|^2 \bar{\Delta}_B / n}$. Together with the bound for the first term, when $\bar{\Delta}_B < \mathcal{M}_\lambda$, the loss difference is bounded by $(c_B + m\gamma c_\alpha) \sqrt{\sum_{l=1}^K |C_l|^2 \bar{\Delta}_B / n}$.

Second, if $\bar{\Delta}_B \geq \mathcal{M}_\lambda$, the cluster assignments may change. Since $0 \leq \sum_{k=1}^m \alpha_{k,\mathcal{X}}^\gamma p_{k,\mathcal{X}}(\mathbf{x}_i, \mathbf{x}_j) \leq 1$ and $\mathbb{I}[g(\mathbf{x}_i) = g(\mathbf{x}_j)] \in \{0, 1\}$, the pairwise loss lies in an interval whose length is at most $\lambda + 1$. Thus, the loss difference is bounded by $\lambda + 1$.

Combining the two cases, for any $\mathbf{x}_i, \mathbf{x}_j$,

$$\begin{aligned} & \sup_{\mathbf{x}_i, \mathbf{x}_j} \left| \ell_\lambda(\mathbf{x}_i, \mathbf{x}_j; A_\lambda(\mathcal{X})) - \ell_\lambda(\mathbf{x}_i, \mathbf{x}_j; A_\lambda(\mathcal{X}^{(r)})) \right| \\ & \leq (c_B + m\gamma c_\alpha) \frac{\sqrt{\sum_{l=1}^K |C_l|^2}}{n} \bar{\Delta}_B + (\lambda + 1) \mathbb{I}[\bar{\Delta}_B \geq \mathcal{M}_\lambda]. \end{aligned} \quad (\text{S45})$$

Taking the supremum over all $\mathbf{x}_i, \mathbf{x}_j$ completes the proof of Theorem 6. $\square$

SUPPLEMENTAL MATERIAL B COMPLEXITY ANALYSIS

In this section, we further discuss the space complexity of OLEC-SB. The indicator matrix $\mathbf{Y} \in \{0, 1\}^{n \times K}$ requires $\mathcal{O}(nK)$ space, and the weight vector $\alpha \in \mathbb{R}^m$ costs $\mathcal{O}(m)$ space. For each base clustering, $\mathbf{B}_k \in \mathbb{R}^{n \times l_k}$ contains exactly one nonzero entry per row and can be stored in a sparse form, requiring $\mathcal{O}(n)$ memory per k and $\mathcal{O}(nm)$ in total. Moreover, to accelerate the update of $\mathbf{Y}$, we store $\mathbf{y}_s^\top \mathbf{B}_k$ and $\mathbf{y}_s^\top \mathbf{1}_n$ for all s and k , which needs $\mathcal{O}(K \sum_{k=1}^m l_k)$ and $\mathcal{O}(K)$ space, respectively. Finally, the term $(\mathbf{b}_k)_i (\mathbf{b}_k)_i^\top$ can be computed in $\mathcal{O}(1)$ time due to the one-sparse structure of $(\mathbf{b}_k)_i$, and thus does not need to be explicitly stored. Therefore, the overall space complexity of OLEC-SB is $\mathcal{O}(nK + nm + K \sum_{k=1}^m l_k)$, which is linear in the number of samples n .

SUPPLEMENTAL MATERIAL C ADDITIONAL EXPERIMENTS

In this section, we provide additional details about the comparative experiments that are omitted in the main paper due to space limitations.

A. Details of Datasets

In the experiments in Section IV, we used 10 benchmark datasets including images, sensor data, and textual information etc. We have summarized the feature information of the datasets in Table S1, and the detailed information is as follows:

- **Maternal Health Risk**¹: The dataset consists of clinical health data from pregnant patients, including parameters like blood pressure and blood sugar to predict risk levels.
- **CNAE-9**²: The dataset contains 1,080 documents categorized into nine distinct classes, representing various economic activities of Brazilian companies.
- **Phishing Websites**³: The dataset consists of a collection of legitimate and phishing website instances. Each website is represented by the set of features which denote, whether the website is legitimate or not.
- **PSDAS**⁴: The dataset contains 4,424 records of students from a higher education institution, covering diverse undergraduate degrees such as technologies, nursing, and management.

¹<https://archive.ics.uci.edu/dataset/863/maternal+health+risk>

²<https://archive.ics.uci.edu/dataset/233/cnae+9>

³<http://archive.ics.uci.edu/dataset/327/phishing+websites>

⁴<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>

TABLE S1
DETAILS OF DIFFERENT DATASETS.

No.	Dataset	#Instance	#Feature	#Class	Scale	Cluster-Size Distributions
D1	Maternal Health Risk	1014	6	3	Medium	Imbalanced
D2	CNAE-9	1080	857	9	Medium	Balanced
D3	Phishing Websites	2456	30	2	Medium	Balanced
D4	PSDAS	4424	36	3	Medium	Imbalanced
D5	TUANDROMD	4464	241	2	Medium	Imbalanced
D6	Gender Gap	4746	18	3	Medium	Imbalanced
D7	COIL100	7200	1024	100	Large	Balanced
D8	Crowdsourced Mapping	10845	29	6	Large	Imbalanced
D9	HTRU2	17898	8	2	Large	Imbalanced
D10	Steel Energy	35040	9	3	Large	Imbalanced

TABLE S2
PERFORMANCE COMPARISON ON DIFFERENT DATASETS IN TERMS OF RUNNING TIME (SEC.) AND ARI (MEAN $\pm$ STD, %). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	Metric	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
PTGP [11]	time↓	0.04	0.20	0.65	3.06	0.36	1.95	3.72	178.95	1601.57	51.33	184.18
	ARI↑	11.1 $\pm$ 5.2	19.1 $\pm$ 1.7	-0.3 $\pm$ 0.2	2.8 $\pm$ 3.2	6.4 $\pm$ 18.4	9.1 $\pm$ 9.4	24.2 $\pm$ 1.4	15.8 $\pm$ 2.2	46.4 $\pm$ 1.3	17.7 $\pm$ 5.4	15.2 $\pm$ 4.8
SEC [29]	time↓	0.00	0.01	0.01	0.03	0.01	0.03	0.79	0.16	0.13	0.47	0.16
	ARI↑	12.5 $\pm$ 2.4	19.2 $\pm$ 3.7	1.6 $\pm$ 6.0	2.2 $\pm$ 2.4	4.8 $\pm$ 30.0	9.1 $\pm$ 8.3	38.3 $\pm$ 1.2	6.7 $\pm$ 16.0	37.2 $\pm$ 29.1	2.3 $\pm$ 7.2	13.4 $\pm$ 10.6
LWEA [6]	time↓	0.25	0.30	1.14	3.31	3.61	3.28	5.51	15.16	53.73	152.54	23.88
	ARI↑	15.8 $\pm$ 0.8	21.4 $\pm$ 3.6	-0.6 $\pm$ 0.0	4.8 $\pm$ 2.7	6.5 $\pm$ 19.1	31.2 $\pm$ 2.1	32.8 $\pm$ 1.2	24.5 $\pm$ 6.8	63.2 $\pm$ 13.1	17.7 $\pm$ 5.4	21.7 $\pm$ 5.5
LWGP [6]	time↓	0.09	0.16	0.19	0.33	0.28	0.36	1.57	0.88	1.38	3.17	0.84
	ARI↑	19.3 $\pm$ 5.7	24.6 $\pm$ 2.2	-0.5 $\pm$ 0.0	1.7 $\pm$ 1.5	-2.7 $\pm$ 2.0	30.6 $\pm$ 0.6	39.5 $\pm$ 0.7	21.1 $\pm$ 6.1	55.7 $\pm$ 0.7	20.0 $\pm$ 0.3	20.9 $\pm$ 2.0
ECPCS-HC [16]	time↓	0.22	0.70	1.19	3.62	3.72	14.21	17.69	45.00	67.77	262.42	41.66
	ARI↑	17.5 $\pm$ 1.8	11.0 $\pm$ 1.3	-0.5 $\pm$ 0.2	3.0 $\pm$ 2.1	-3.7 $\pm$ 3.3	30.4 $\pm$ 0.4	28.5 $\pm$ 1.8	15.2 $\pm$ 1.8	58.4 $\pm$ 9.6	5.8 $\pm$ 9.9	16.5 $\pm$ 3.2
ECPCS-MC [16]	time↓	0.09	0.17	0.32	0.73	0.73	1.21	3.87	5.08	4.59	12.55	2.93
	ARI↑	20.4 $\pm$ 6.1	11.9 $\pm$ 1.5	-0.3 $\pm$ 0.2	3.9 $\pm$ 3.2	-2.0 $\pm$ 0.4	30.8 $\pm$ 0.1	35.2 $\pm$ 1.2	13.5 $\pm$ 2.3	52.3 $\pm$ 1.6	3.4 $\pm$ 11.2	16.9 $\pm$ 2.8
U-SENC [12]	time↓	<u>0.02</u>	<u>0.04</u>	<u>0.06</u>	<u>0.13</u>	<u>0.10</u>	<u>0.13</u>	<u>0.85</u>	<u>0.47</u>	<u>0.85</u>	<u>1.71</u>	<u>0.43</u>
	ARI↑	13.5 $\pm$ 8.6	21.5 $\pm$ 1.2	-0.4 $\pm$ 0.2	4.1 $\pm$ 2.7	1.8 $\pm$ 9.2	30.9 $\pm$ 2.2	40.5 $\pm$ 1.0	15.3 $\pm$ 3.2	52.8 $\pm$ 1.2	13.0 $\pm$ 11.5	19.3 $\pm$ 4.1
ECCMS [7]	time↓	9.35	11.99	67.75	340.66	424.44	401.62	1422.21	4245.39	N/A	N/A	N/A
	ARI↑	18.9 $\pm$ 2.8	20.7 $\pm$ 3.6	-0.6 $\pm$ 0.1	4.7 $\pm$ 3.1	4.0 $\pm$ 19.0	28.2 $\pm$ 3.8	26.1 $\pm$ 3.3	24.0 $\pm$ 3.6	N/A	N/A	N/A
FSEC [21]	time↓	0.27	0.50	0.72	13.94	13.93	1.49	3.05	5.43	22.54	21.20	8.31
	ARI↑	9.9 $\pm$ 6.8	31.2 $\pm$ 2.9	1.3 $\pm$ 2.0	3.6 $\pm$ 1.0	75.5 $\pm$ 2.9	25.8 $\pm$ 9.6	36.8 $\pm$ 0.8	17.3 $\pm$ 7.5	39.3 $\pm$ 4.8	9.3 $\pm$ 9.6	25.0 $\pm$ 4.8
YACHT [20]	time↓	1.56	1.14	3.72	20.18	12.83	10.18	23.18	94.48	81.06	178.68	36.70
	ARI↑	12.6 $\pm$ 4.8	30.1 $\pm$ 3.2	28.3 $\pm$ 14.2	5.3 $\pm$ 3.8	18.6 $\pm$ 14.5	23.1 $\pm$ 11.1	44.3 $\pm$ 1.5	18.6 $\pm$ 8.4	-4.4 $\pm$ 2.3	12.8 $\pm$ 8.6	18.9 $\pm$ 7.3
SDGCA [5]	time↓	27.87	25.13	121.11	601.67	1283.63	1250.41	4525.16	N/A	N/A	N/A	N/A
	ARI↑	19.1 $\pm$ 5.9	21.2 $\pm$ 2.3	-0.6 $\pm$ 0.0	5.2 $\pm$ 2.7	-2.0 $\pm$ 0.2	29.2 $\pm$ 1.7	25.8 $\pm$ 4.3	N/A	N/A	N/A	N/A
GPEC [8]	time↓	62.97	100.08	231.57	777.87	942.59	861.61	3999.35	N/A	N/A	N/A	N/A
	ARI↑	8.8 $\pm$ 3.3	26.4 $\pm$ 3.6	33.7 $\pm$ 9.2	3.1 $\pm$ 1.9	-7.6 $\pm$ 2.9	22.1 $\pm$ 9.7	41.7 $\pm$ 1.7	N/A	N/A	N/A	N/A
FCE [9]	time↓	1.27	1.34	3.60	10.88	7.93	9.92	242.65	55.48	90.79	393.27	81.71
	ARI↑	4.7 $\pm$ 2.1	32.1 $\pm$ 4.5	1.2 $\pm$ 1.2	1.2 $\pm$ 0.9	0.1 $\pm$ 0.1	11.3 $\pm$ 5.8	36.1 $\pm$ 1.6	1.3 $\pm$ 0.2	0.0 $\pm$ 0.0	1.3 $\pm$ 1.1	8.9 $\pm$ 1.7
TBAWEC [36]	time↓	30.47	66.13	249.51	2461.23	1378.39	3077.75	N/A	N/A	N/A	N/A	N/A
	ARI↑	11.5 $\pm$ 2.5	35.3 $\pm$ 3.3	54.9 $\pm$ 29.6	5.6 $\pm$ 3.4	13.6 $\pm$ 5.4	18.5 $\pm$ 1.4	N/A	N/A	N/A	N/A	N/A
OLEC-SB (Ours)	time↓	0.07	0.33	0.86	1.16	1.09	1.00	17.32	6.28	12.86	244.77	28.57
	ARI↑	22.8 $\pm$ 4.8	40.7 $\pm$ 3.4	60.3 $\pm$ 13.3	12.0 $\pm$ 3.4	80.9 $\pm$ 2.2	38.7 $\pm$ 2.0	44.0 $\pm$ 1.5	32.4 $\pm$ 7.7	79.2 $\pm$ 4.1	27.6 $\pm$ 2.4	43.9 $\pm$ 4.5

N/A indicates excessive runtime (mean execution time of 10 runs > 5000 s) or insufficient memory.

- **TUANDROMD**⁵: The dataset contains 4,464 instances of Android applications, categorized into malware and goodware to support security research.
- **Gender Gap**⁶: The dataset consists of 475 samples representing individual editors from the Spanish Wikipedia community.
- **COIL100**⁷: The dataset contains 7,200 color images of

100 different objects, with 72 images per object taken from various angles.

- **Crowdsourced Mapping**⁸: The dataset includes 29 features, such as NDVI (Normalized Difference Vegetation Index) values and other spectral data extracted from multiple timestamps across different seasons.
- **HTRU2**⁹: The dataset contains 17,898 samples, each representing a candidate for a pulsar signal identified

⁵<https://archive.ics.uci.edu/dataset/813/tunadromd>

⁶<https://archive.ics.uci.edu/dataset/852/gender+gap+in+spanish+wp>

⁷<https://www.cs.columbia.edu/CAVE/software/softlib/coil-100.php>

⁸<https://archive.ics.uci.edu/dataset/400/crowdsourced+mapping>

⁹<https://archive.ics.uci.edu/dataset/372/htru2>

TABLE S3

PERFORMANCE COMPARISON ON DIFFERENT DATASETS IN TERMS OF RUNNING TIME (SEC.) AND F-SCORE (MEAN $\pm$ STD, %). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	Metric	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
PTGP [11]	time↓	0.04	0.20	0.65	3.06	0.36	1.95	3.72	178.95	1601.57	51.33	184.18
	F-score↑	43.1 $\pm$ 4.1	31.5 $\pm$ 1.1	57.2 $\pm$ 2.1	39.9 $\pm$ 1.2	76.3 $\pm$ 7.0	53.4 $\pm$ 2.8	25.3 $\pm$ 1.4	38.8 $\pm$ 2.5	87.4 $\pm$ 0.6	48.4 $\pm$ 3.1	50.1 $\pm$ 2.6
SEC [29]	time↓	0.00	0.01	0.01	0.03	0.01	0.03	0.79	0.16	0.13	0.47	0.16
	F-score↑	49.8 $\pm$ 3.4	31.7 $\pm$ 2.6	60.6 $\pm$ 2.3	<u>50.4</u> $\pm$ 3.4	74.0 $\pm$ 8.6	57.1 $\pm$ 3.8	39.1 $\pm$ 1.2	47.4 $\pm$ 10.6	91.3 $\pm$ 5.9	<u>53.6</u> $\pm$ 2.9	<u>55.5</u> $\pm$ 4.5
LWEA [6]	time↓	0.25	0.30	1.14	3.31	3.61	3.28	5.51	15.16	53.73	152.54	23.88
	F-score↑	49.8 $\pm$ 1.8	33.6 $\pm$ 2.3	61.5 $\pm$ 0.5	41.8 $\pm$ 2.4	<u>80.0</u> $\pm$ 2.4	61.9 $\pm$ 0.3	33.7 $\pm$ 1.2	<u>47.9</u> $\pm$ 7.4	<u>92.9</u> $\pm$ 3.5	48.9 $\pm$ 2.3	55.2 $\pm$ 2.4
LWGP [6]	time↓	0.09	0.16	0.19	0.33	0.28	0.36	1.57	0.88	1.38	3.17	0.84
	F-score↑	50.2 $\pm$ 4.4	35.8 $\pm$ 1.7	61.7 $\pm$ 0.0	40.6 $\pm$ 1.5	77.9 $\pm$ 3.3	61.7 $\pm$ 0.4	40.2 $\pm$ 0.7	45.4 $\pm$ 5.6	90.9 $\pm$ 0.2	49.9 $\pm$ 0.1	55.4 $\pm$ 1.8
ECPCS-HC [16]	time↓	0.22	0.70	1.19	3.62	3.72	14.21	17.69	45.00	67.77	262.42	41.66
	F-score↑	51.2 $\pm$ 0.6	27.0 $\pm$ 0.8	60.5 $\pm$ 2.6	40.9 $\pm$ 1.8	77.0 $\pm$ 4.1	61.8 $\pm$ 0.2	29.5 $\pm$ 1.7	41.1 $\pm$ 1.1	91.5 $\pm$ 2.8	45.9 $\pm$ 2.8	52.6 $\pm$ 1.9
ECPCS-MC [16]	time↓	0.09	0.17	0.32	0.73	0.73	1.21	3.87	5.08	4.59	12.55	2.93
	F-score↑	52.3 $\pm$ 1.8	27.6 $\pm$ 0.9	57.7 $\pm$ 3.4	41.0 $\pm$ 1.2	79.1 $\pm$ 0.5	61.7 $\pm$ 0.0	36.1 $\pm$ 1.1	38.3 $\pm$ 2.7	89.7 $\pm$ 0.6	45.4 $\pm$ 3.0	52.9 $\pm$ 1.5
U-SENC [12]	time↓	0.02	0.04	0.06	0.13	0.10	0.13	0.85	0.47	0.85	1.71	0.43
	F-score↑	46.2 $\pm$ 5.9	33.0 $\pm$ 0.8	58.9 $\pm$ 3.0	40.2 $\pm$ 0.9	73.8 $\pm$ 7.3	60.3 $\pm$ 2.2	41.2 $\pm$ 1.0	39.1 $\pm$ 2.8	89.9 $\pm$ 0.4	48.1 $\pm$ 3.1	53.1 $\pm$ 2.7
ECCMS [7]	time↓	9.35	11.99	67.75	340.66	424.44	401.62	1422.21	4245.39	N/A	N/A	N/A
	F-score↑	51.0 $\pm$ 1.9	33.1 $\pm$ 2.4	61.2 $\pm$ 1.7	41.8 $\pm$ 4.0	79.9 $\pm$ 3.0	<u>62.3</u> $\pm$ 2.2	27.2 $\pm$ 3.2	47.4 $\pm$ 4.1	N/A	N/A	N/A
FSEC [21]	time↓	0.27	0.50	0.72	13.94	13.93	1.49	3.05	5.43	22.54	21.20	8.31
	F-score↑	44.3 $\pm$ 4.9	39.3 $\pm$ 2.5	55.6 $\pm$ 3.4	44.6 $\pm$ 1.1	92.2 $\pm$ 0.9	57.3 $\pm$ 5.4	37.6 $\pm$ 0.7	41.5 $\pm$ 7.3	84.4 $\pm$ 2.2	48.2 $\pm$ 1.7	54.5 $\pm$ 3.0
YACHT [20]	time↓	1.56	1.14	3.72	20.18	12.83	10.18	23.18	94.48	81.06	178.68	36.70
	F-score↑	43.8 $\pm$ 3.3	38.9 $\pm$ 2.7	65.8 $\pm$ 7.6	40.8 $\pm$ 2.5	66.8 $\pm$ 5.0	57.7 $\pm$ 5.5	44.9 $\pm$ 1.5	43.6 $\pm$ 6.9	65.4 $\pm$ 2.9	48.1 $\pm$ 4.2	51.5 $\pm$ 4.2
SDGCA [5]	time↓	27.87	25.13	121.11	601.67	1283.63	1250.41	4525.16	N/A	N/A	N/A	N/A
	F-score↑	51.2 $\pm$ 3.8	38.4 $\pm$ 1.6	61.7 $\pm$ 0.0	41.3 $\pm$ 2.5	79.0 $\pm$ 0.1	60.7 $\pm$ 1.4	27.0 $\pm$ 4.1	N/A	N/A	N/A	N/A
GPEC [8]	time↓	62.97	100.08	231.57	777.87	942.59	861.61	3999.35	N/A	N/A	N/A	N/A
	F-score↑	41.3 $\pm$ 0.9	36.2 $\pm$ 2.8	67.8 $\pm$ 5.0	39.1 $\pm$ 0.8	62.3 $\pm$ 1.7	54.7 $\pm$ 5.2	42.4 $\pm$ 1.7	N/A	N/A	N/A	N/A
FCE [9]	time↓	1.27	1.34	3.60	10.88	7.93	9.92	242.65	55.48	90.79	393.27	81.71
	F-score↑	36.8 $\pm$ 1.4	39.6 $\pm$ 4.0	50.8 $\pm$ 0.6	36.5 $\pm$ 0.5	57.6 $\pm$ 0.0	45.8 $\pm$ 3.5	36.8 $\pm$ 1.5	26.1 $\pm$ 0.2	62.5 $\pm$ 0.0	36.6 $\pm$ 0.7	42.9 $\pm$ 1.3
TBAWEC [36]	time↓	30.47	66.13	249.51	2461.23	1378.39	3077.75	N/A	N/A	N/A	N/A	N/A
	F-score↑	41.3 $\pm$ 1.7	<u>44.6</u> $\pm$ 2.9	<u>77.6</u> $\pm$ 14.7	39.3 $\pm$ 2.2	63.0 $\pm$ 2.3	50.1 $\pm$ 0.8	N/A	N/A	N/A	N/A	N/A
OLEC-SB (Ours)	time↓	0.07	0.33	0.86	1.16	1.09	1.00	17.32	6.28	12.86	244.77	28.57
	F-score↑	<u>51.8</u> $\pm$ 3.2	47.3 $\pm$ 3.0	80.2 $\pm$ 6.6	51.0 $\pm$ 2.2	93.4 $\pm$ 0.8	63.4 $\pm$ 1.4	<u>44.6</u> $\pm$ 1.5	70.4 $\pm$ 2.9	96.7 $\pm$ 0.9	60.5 $\pm$ 1.0	65.9 $\pm$ 2.4

N/A indicates excessive runtime (mean execution time of 10 runs $>$ 5000 s) or insufficient memory.

TABLE S4

THE AVERAGE RANK AR_i OF EACH METHOD ON DATASETS D1-D6.

	PTGP	SEC	LWEA	LWGP	ECPCS-HC	ECPCS-MC	U-SENC	ECCMS	FSEC	YACHT	SDGCA	GPEC	FCE	TBAWEC	OLEC-SB
ACC	12.33	8.83	6.67	7.33	8.33	7.50	9.67	6.83	5.83	7.67	6.33	10.00	11.83	8.83	2.00
ARI	11.00	10.17	7.00	9.00	10.50	8.00	7.50	8.50	7.17	5.67	8.67	9.83	10.50	5.50	1.00
F-score	12.00	8.50	5.33	6.83	8.00	7.17	10.00	6.67	7.17	8.33	5.83	10.33	13.00	9.67	1.17

during the High Time Resolution Universe Survey.

- **Steel Energy**¹⁰: The dataset consists of 35,040 instances recorded at 15-minute intervals over a one-year period from a steel plant in South Korea.

B. Compared Methods

We compared our method with the following methods.

- **PTGP** [11]: this method builds a microcluster-cluster bipartite graph, assigns weights using dense similarities from random walk probability trajectories on a sparse graph, and then performs graph partitioning.
- **SEC** [29]: this method performs spectral clustering on the co-association matrix and accelerates it via the theoretical equivalence to weighted k -means.

- **LWEA and LWGP** [6]: these methods estimate cluster reliability via the ensemble-driven cluster index (ECI). LWEA builds a locally weighted co-association matrix and applies hierarchical clustering, while LWGP integrates ECI into sample-cluster link weights to construct a locally weighted bipartite graph for partitioning.
- **ECPCS-HC and ECPCS-MC** [16]: these methods enhance co-association matrix by exploiting multiscale cluster similarities derived from random walks. ECPCS-HC transfers cluster-level similarities to the sample level to build an enhanced co-association matrix and applies hierarchical clustering, while ECPCS-MC partitions the cluster similarity graph into meta-clusters via normalized-cut and assigns samples by majority voting.
- **U-SENC** [12]: this method integrates multiple ultra-scalable spectral clustering to generate the ensemble and

¹⁰<https://archive.ics.uci.edu/dataset/851/steel+industry+energy+consumption>

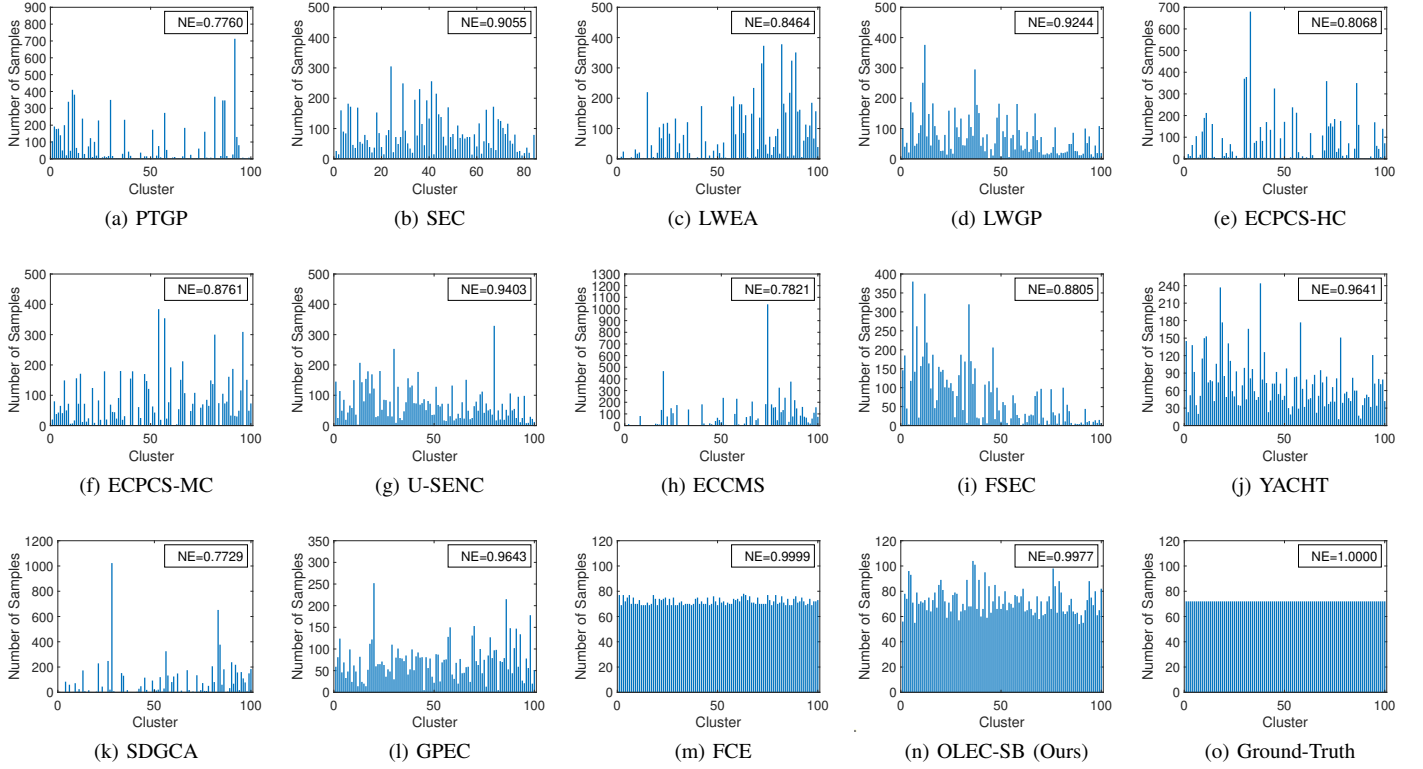


Fig. S1. Distribution of samples in each cluster on the D7 dataset in a single experimental run.

TABLE S5
SUMMARY OF NOTATIONS USED IN OLEC-SB.

Notation	Description
n, d, m, K, l_k	Number of samples, feature dimension, base clusterings, final clusters, and clusters in the k -th base clustering.
$\text{Ind}, \mathbf{I}, \mathbf{1}_n, \mathbf{1}_K, \mathbf{1}_m$	Feasible set of discrete indicator matrices, identity matrix, all-one vectors of lengths n , K , and m .
$\ \cdot\ _F, \text{tr}(\cdot), (\cdot)^\top, \succeq$	Frobenius norm, trace operator, transpose operator, and element-wise nonnegativity.
$\mathcal{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n], \mathbf{x}_i, \mathcal{X}^{(r)}$	Input dataset, the i -th sample, and perturbed dataset obtained by replacing only the r -th sample in $\mathcal{X}$.
$\Pi = \{\pi_1, \dots, \pi_m\}, \pi_k$	Set of base clusterings and the k -th base clustering.
$\mathcal{C} = \{\mathcal{C}_1, \dots, \mathcal{C}_K\}, \mathcal{C}_l, \bar{\mathcal{C}}_l, \mathcal{C}_l $	Consensus partition, the l -th cluster of $\mathcal{C}$, the complement of $\mathcal{C}_l$, and the size of $\mathcal{C}_l$.
$\mathbf{A}_k \in \{0, 1\}^{n \times K}, (\mathbf{a}_k)_i$	Indicator matrix of the k -th base clustering, and the i -th row of $\mathbf{A}_k$.
$\Delta_k = \mathbf{A}_k^\top \mathbf{A}_k$	Diagonal matrix recording the sizes of clusters in π_k .
$\mathbf{B}_k = \mathbf{A}_k \Delta_k^{-1/2}, (\mathbf{b}_k)_i$	Normalized indicator matrix of the k -th base clustering, and the i -th row of $\mathbf{B}_k$.
$\mathbf{W}_k = \mathbf{B}_k \mathbf{B}_k^\top, w_{ij}^k$	Normalized co-association matrix of the k -th base clustering, and the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$.
$\mathbf{D}_k = \text{diag}(d_1^k, \dots, d_n^k)$	Degree matrix of $\mathbf{W}_k$. Under the normalized construction, $\mathbf{D}_k = \mathbf{I}$.
$\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_K] \in \{0, 1\}^{n \times K}, \mathbf{y}_l$	Discrete indicator matrix of the final clustering result, and the l -th column of $\mathbf{Y}$.
$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_m]^\top, \alpha_k$	Weight vector of base clusterings, and weight of π_k .
$\mathbf{Y}^{(s)}, \mathbf{Y}^{(0)}, p, q, s$	Candidate matrix, auxiliary matrix, current cluster index, updated cluster index, and candidate cluster index.
$\boldsymbol{\delta}_i, \varphi(s)$	One-hot vector with the i -th entry equal to 1, and the objective difference for assigning $\mathbf{x}_i$ to cluster s .
$\mathbf{y}_s^\top \mathbf{B}_k, \mathbf{y}_s^\top \mathbf{1}_n, (\mathbf{b}_k)_i (\mathbf{b}_k)_i^\top$	Intermediate quantities stored to accelerate the computation of $\varphi(s)$ in fast coordinate descent.
$\mathbf{y}_p^\top \mathbf{B}_k, \mathbf{y}_q^\top \mathbf{B}_k, \mathbf{y}_p^\top \mathbf{1}_n, \mathbf{y}_q^\top \mathbf{1}_n$	Cached variables updated when $\mathbf{x}_i$ is reassigned from cluster p to cluster q .
$\mathcal{C}_s^{-i}, \mathcal{C}_s^{-i} , \bar{a}_{is}$	The s -th cluster after temporarily removing $\mathbf{x}_i$, the size of $\mathcal{C}_s^{-i}$, and average similarity between $\mathbf{x}_i$ and $\mathcal{C}_s^{-i}$.
$A_\lambda(\mathcal{X}) = (g_\lambda, \boldsymbol{\alpha}_\lambda, \{\mathbf{B}_{k,\lambda}\}_{k=1}^m)$	Output of OLEC-SB on $\mathcal{X}$, i.e. the assignment function, weight vector, and normalized indicator matrices.
$p_{k,\lambda}(\mathbf{x}_i, \mathbf{x}_j), \ell_\lambda(\mathbf{x}_i, \mathbf{x}_j; A_\lambda(\mathcal{X}))$	Pairwise normalized co-association value and the pairwise loss of OLEC-SB.
$\Delta \mathbf{B}_k = \mathbf{B}_{k,\lambda^{(r)}} - \mathbf{B}_{k,\lambda}, M_\lambda, \xi$	Perturbation of the normalized indicator matrix, reassignment margin, and pairwise uniform stability bound.
$\mathbb{P}_n, \mathbb{P}$	Empirical distribution and underlying data distribution.
$R_\lambda(A_\lambda(\mathcal{X}), \mathbb{P}_n), R_\lambda(A_\lambda(\mathcal{X}), \mathbb{P})$	Empirical pairwise risk and expected pairwise risk.

derives the consensus by partitioning a sample-cluster bipartite graph.

- **ECCMS** [7]: this method extracts high-confidence information from base clusterings to form a sparse matrix and employs a self-enhancement mechanism to propagate reliable information while denoising the co-association matrix.
- **FSEC** [21]: this method constructs an anchor graph via fast k -nearest neighbor approximation, generates base clusterings using repeated k -means on spectral embeddings from a single singular value decomposition, and applies transfer cut on the resulting bipartite graph.
- **YACHT** [20]: this method converts base clusterings into a hypergraph using random walks, reconstructs it via matrix factorization with hypergraph regularization to learn anchors and anchor similarities, and derives the consensus from the learned anchor similarity matrix.
- **SDGCA** [5]: this method incorporates cluster-size information via normalized ensemble entropy to construct a weighted co-association matrix, explores high-order proximity using random walks, and refines similarity and dissimilarity matrices through adversarial learning for hierarchical clustering.
- **GPEC** [8]: this method derives a bias-diversity decomposition from generalization theory, approximates the expected co-association matrix using high-confidence second-order similarities, and learns adaptive weights via min-max optimization followed by spectral embedding and k -means.
- **FCE** [9]: this method introduces a regularization term measuring fairness and cluster capacity equality and integrates it into an ensemble clustering framework optimized iteratively.
- **TBAWEC** [36]: this method formulates ensemble clustering as an optimization problem with normalized similarity matrices and exclusive lasso regularization, and jointly learns adaptive weights and discrete cluster indicators via alternating optimization.

C. Clustering Performance Comparison

In the supplemental material, we further present the results in terms of ARI and F-score. From Table S2, OLEC-SB achieves the best ARI on the majority of datasets, except for D7, where it attains the second-best result. From Table S3 OLEC-SB also achieves the best F-score on most datasets, with second-best performance on D1 and D7.

D. Balance Performance

Figure S1 presents the cluster-size distribution on the D7 dataset for different ensemble clustering methods.

E. Statistical Analysis

Friedman testing [41] is conducted to examine the statistical significance of clustering performance *w.r.t.* fifteen methods in terms of ACC, ARI and F-score. Under the null hypothesis, all methods are presumed to be equivalent; thus, rejecting the null

hypothesis indicates statistically significant differences among the compared methods. The average rank AR_i is calculated by

$$AR_i = \frac{1}{N_F} \sum_j r_{ij}, \quad (\text{S46})$$

where r_{ij} denotes the rank of the i -th method on the j -th dataset. The average rank of the compared methods on ACC, ARI and F-score are summarized in Table S4. The τ_F of Friedman testing is calculated using Eqs. (S47) and (S48):

$$\tau_F = \frac{(N_F - 1)\tau_{\chi^2}}{N_F(K_F - 1) - \tau_{\chi^2}}, \quad (\text{S47})$$

$$\tau_{\chi^2} = \frac{12N_F}{K_F(K_F + 1)} \left(\sum_{i=1}^{K_F} AR_i^2 - \frac{K_F(K_F + 1)^2}{4} \right), \quad (\text{S48})$$

where K_F denotes the number of methods and N_F denotes the number of datasets. In our setting, $K_F = 15$ methods are evaluated on $N_F = 6$ datasets (D1-D6). The Friedman statistic τ_F follows an F distribution with degrees of freedom $(K_F - 1)$ and $(K_F - 1)(N_F - 1)$. At the 95% confidence level, the critical value is $F_{0.05}((K_F - 1), (K_F - 1)(N_F - 1)) = F_{0.05}((15 - 1), (15 - 1)(6 - 1)) = 1.8357$. Since this critical value is much smaller than the computed τ_F values (2.3362 for ACC, 2.5224 for ARI, 3.5685 for F-score), we conclude that the differences among the fifteen methods are statistically significant at the 5% level.

SUPPLEMENTAL MATERIAL D NOTATION SUMMARY

In this section, we summarize the main notations used in OLEC-SB and the theoretical analysis in Table S5.